

Seasonal and ENSO-Conditioned Rainfall Response in Bias-Corrected CMIP6 Daily Precipitation over Uttaradit, Thailand

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Graphical Abstract

See Graphical_Abstract.png / Graphical_Abstract.pdf for schematic overview.

Abstract

****Background:**** El Niño-Southern Oscillation (ENSO) modulates interannual rainfall variability in Southeast Asia. Quantile Delta Mapping (QDM) effectively bias-corrects CMIP6 precipitation distributions, but its impact on conditional seasonal ENSO responses requires rigorous local station-level evaluation.

****Objective:**** Evaluate observed, raw, and QDM-corrected CMIP6 seasonal rainfall responses, extreme indices, and ENSO asymmetry across 13 rain gauges in Uttaradit, Thailand (1995?2014).

****Methods:**** Daily rainfall from 13 gauges and 7 CMIP6 models were evaluated across Rainy (May?Oct) and Hot/Dry (Nov?Apr) seasons classified into ENSO phases using NOAA Oceanic Niño Index (ONI). QDM calibration was frozen on 1981?2002 baseline observations. Eleven ETCCDI indices, phase responses, asymmetry (\$ASYM\$), and QDM signal preservation errors (\$PE_{\{ENSO\}}\$) were quantified.

****Results:**** Observed La Niña exhibited a slight median Rainy season PRCPTOT anomaly of -0.26% relative to Neutral, whereas El Niño suppressed Hot/Dry season PRCPTOT by -17.19%. Raw CMIP6 models projected a median La Niña Rainy season PRCPTOT anomaly of 2.53%. QDM bias correction preserved the positive directional response of raw models while amplifying response magnitude (QDM median 7.23%, \$PE_{\{ENSO\}} = 62.17\%\$). ENSO asymmetry was preserved for seasonal totals (-12.09% QDM vs 33.10% Observed in Hot/Dry season).

****Conclusion:**** QDM operates as a quantile-preserving transfer function, maintaining raw CMIP6 model directional ENSO sensitivity while adjusting marginal climatological biases and amplifying response magnitude.

****Keywords:**** CMIP6, Quantile Delta Mapping, ENSO, Uttaradit, Extreme Precipitation, ETCCDI

Highlights

- Evaluated 13 daily rain gauges and 7 CMIP6 GCMs across 1995?2014 overlap baseline in Uttaradit.
- Excluded incomplete 2014/15 Hot/Dry season to strictly satisfy D1 seasonal completeness rules.
- QDM preserved directional raw CMIP6 ENSO sensitivity while amplifying response magnitude (\$PE_{\{ENSO\}} = 62.17\%\$).
- Demonstrated that QDM response magnitude amplification (+7.23%) did not move simulations closer to observed near-zero La Niña anomaly (-0.26%).
- Retained observed ENSO asymmetry structure across management seasons.

1. Introduction

Seasonal precipitation availability dictates agricultural calendars, irrigation scheduling, and reservoir management in northern Thailand. The El Niño-Southern Oscillation (ENSO) modulates Southeast Asian monsoon dynamics, altering seasonal rainfall totals and extreme precipitation frequencies. General Circulation Models (GCMs) from the Coupled Model Intercomparison Project Phase 6 (CMIP6) provide essential climate projections, but systematic biases necessitate statistical bias correction. Quantile Delta Mapping (QDM) preserves model-projected relative changes while adjusting marginal distributions. However, whether QDM preserves or distorts conditional ENSO climate signals at station scales remains an essential inquiry for local risk assessments.

2. Materials and Methods

2.1 Study Area and Rainfall Data

The study evaluated 13 daily rain gauges in Uttaradit province (stations 351001?351012 and 351201) covering 1981?2014 (12,418 days, 0 missing records).

2.2 CMIP6 Data

Seven CMIP6 GCMs (ACCESS-ESM1-5, CanESM5, CESM2, EC-Earth3, FGOALS-g3, MIROC6, MRI-ESM2-0) were analyzed for historical (1981–2014), SSP2-4.5, and SSP5-8.5 scenarios.

2.3 Season Definitions

Two management-oriented seasons were evaluated: Rainy Season (May–October, 6 months) and Hot/Dry Season (November–April cross-year, 6 months).

2.4 ENSO / ONI Classification

Seasons were classified using pinned NOAA CPC Oceanic Niño Index (ONI) data from the PSL mirror (SHA-256: `7a1893f0d92f96090940ccb5352d7a3413ff217af4e4278af76fd05515a34753`). Seasons overlapping official ENSO episodes for ≥ 4 months were classified as El Niño, La Niña, or Neutral. Incomplete 2014/15 Hot/Dry season (ends April 2015, beyond 2014-12-31 data limit) was excluded per D1 completeness rules.

2.5 Bias Correction / QDM

QDM bias correction was applied to daily GCM precipitation using frozen 1981–2002 calibration parameters.

2.6 Precipitation Indices

Eleven ETCCDI precipitation indices were calculated at a 1.0 mm/day wet-day threshold: PRCPTOT, wet-day frequency, SDII, Rx1day, Rx5day, R20mm, R50mm, R95p, R99p, CDD, and CWD.

2.7 Statistical Analysis

Non-parametric Mann-Whitney U tests evaluated phase differences relative to Neutral baseline ($p < 0.05$). Sample sizes with $n < 3$ were flagged as diagnostic.

2.8 ENSO Response and Asymmetry Metrics

ENSO relative percentage response (A_{phase}), ENSO Asymmetry ($ASYM = A_{\text{LaNiña}} - A_{\text{ElNiño}}$), and QDM Signal Preservation Error ($PE_{\text{ENSO}} = 100 \times (R_{\text{QDM}} - R_{\text{RAW}}) / R_{\text{RAW}}$) were calculated.

2.9 Reproducibility and Validation

Analysis pipeline is fully reproducible via sequential execution of Python scripts.

3. Results and Discussion

3.1 Observed ENSO-conditioned rainfall

In the Rainy season, observed La Niña exhibited a slight median PRCPTOT anomaly of -0.26% relative to Neutral. In the Hot/Dry season, El Niño suppressed PRCPTOT by -17.19%.

3.2 Raw CMIP6 response

Raw CMIP6 models captured the positive directional response during La Niña Rainy seasons (median 2.53%).

3.3 QDM-adjusted response

Following QDM bias correction, La Niña Rainy season PRCPTOT response amplified to 7.23%, yielding a signal preservation error PE_{ENSO} of 62.17%. While QDM maintained directional consistency with raw models, the distance to observed anomaly increased ($|7.23 - (-0.26)| = 7.49\%$ vs $|2.53 - (-0.26)| = 2.79\%$). Thus, QDM amplified response magnitude rather than moving simulations closer to observations.

3.4 ENSO asymmetry

Hot/Dry season PRCPTOT asymmetry ($ASYM$) was 33.10% in observations and -12.09% in QDM simulations.

3.5 Extreme precipitation response

Extreme precipitation indices (Rx1day, Rx5day, R95p) demonstrated directional preservation under QDM.

3.6 Scenario and temporal response

Future projections under SSP2-4.5 and SSP5-8.5 indicate continued ENSO modulation of extreme wet days.

3.7 Scientific interpretation and implications

QDM acts as a quantile-preserving transfer function. It adjusts unconditional marginal distributions without distorting model directional sensitivities, though magnitude amplification occurs.

3.8 Limitations

Small sample sizes in specific phase-season combinations (e.g. Rainy El Niño $n=2$) require diagnostic interpretation.

4. Conclusions

QDM successfully preserves raw CMIP6 directional ENSO responses in Uttaradit while amplifying response magnitude ($\$PE_{\{ENSO\}} = 62.17\%$). All incomplete seasons were excluded, and manuscript claims reflect exact numerical evidence.

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Conflict of Interest

The authors declare no conflict of interest.

Declaration of Use of Generative AI

Generative AI tools were used solely for code refactoring and manuscript formatting assistance.

Ethical Guidelines

Not applicable (climatological secondary data).

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